

Cluster Naming Evaluation Report (Version 3)

OWNER vs. Span2Type: Cluster Naming Comparison

BERTScore Evaluation across Five CrossNER Domains

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Abstract

This report compares four cluster-naming methods across five CrossNER domains (AI, Literature, Music, Politics, Science), evaluated using BERTScore (RoBERTa-large) against ground-truth entity type labels. Two methods belong to **OWNER**: BERT-based naming (MLM top-1 vote) and LLM-based naming (Llama-3.1-8B via Ollama, Figure 3 prompt, $n=16$). Two methods belong to **Span2Type**: MLM-based naming (same BERT voting strategy) and Ollama-based naming (Llama-3.1-8B with richer context prompting). OWNER-BERT and Span2Type-MLM produce **identical** F_1 scores (0.966) since they share the same underlying approach. Span2Type-Ollama achieves a higher average F_1 of **0.952** compared to OWNER-LLM’s **0.944**, driven by more descriptive multi-word names.

1 Introduction

Unsupervised NER systems such as OWNER [1] group extracted entities into clusters identified only by integer IDs. Automatic cluster naming assigns human-readable type labels without manual annotation. This report evaluates and compares four naming strategies:

- **OWNER-BERT**: for each entity in a cluster, fill the prompt ‘‘{sentence} {entity} is a [MASK].’’ and take the top-1 MLM prediction; the most frequent token across the cluster is the name.
- **OWNER-LLM**: sample $n=16$ entities per cluster, query Llama-3.1-8B with the Figure 3 prompt (temperature = 0), use the response as the name.
- **Span2Type-MLM**: same BERT voting strategy as OWNER-BERT, applied to Span2Type cluster assignments.
- **Span2Type-Ollama**: Llama-3.1-8B with richer context (full example sentences included in the prompt), producing longer and more specific multi-word names.

All methods are scored against ground-truth labels using BERTScore [2] (RoBERTa-large backbone).

2 Experimental Setup

2.1 MLflow Run IDs

Table 1: MLflow run IDs for OWNER and Span2Type experiments.

Domain	OWNER Run ID	Span2Type Run ID
AI	ea6e40f8b9ed4aa385956217b939eb6e	6b105207f3084468b2b79f37841ae2b2
Literature	fb1215f7b8e642ee99d43782427432f0	29e11a34c7214005b0d96bc0520515be
Music	2c886b9b2f104bd4ad79ed1bf3dd13c5	56ad65e38f4e464f85acc934340692dc
Politics	9c02f083100c41b3b052373d4e8e916d	610c691d79904252b375511432bbbe7b
Science	f72db65d19f243a6aedda8687a538896	efcfb115a1cf47798145e9ee97b01755

2.2 Ground-Truth Alignment

For OWNER runs, the GT label per cluster comes from `et_test_cluster_naming_gt.json` (column-wise argmax of the confusion matrix). For Span2Type runs (which do not store a GT file), the same argmax procedure is applied directly to the confusion matrix tensor (`et_test_confusion-matrix.pt`) and its metadata.

2.3 Cluster Statistics

Table 2: Number of aligned clusters per domain and run type.

Domain	True k	OWNER clusters	Span2Type clusters
AI	14	16	16
Literature	12	14	14
Music	13	20	20
Politics	9	22	22
Science	17	20	20
Total	65	92	92

3 Results

3.1 BERTScore Comparison

Table 3 presents P, R, F_1 for all four methods. The best F_1 per domain is shown in **bold**.

Table 3: BERTScore (P / R / F_1) for four cluster-naming methods across five CrossNER domains. Best F_1 per domain in **bold**.

Domain	OWNER-BERT			OWNER-LLM			S2T-MLM			S2T-Ollama		
	P	R	F_1	P	R	F_1	P	R	F_1	P	R	F_1
AI	0.992	0.986	0.989	0.915	0.945	0.929	0.992	0.986	0.989	0.934	0.956	0.944
Literature	0.974	0.975	0.974	0.950	0.960	0.954	0.974	0.975	0.974	0.969	0.970	0.969
Music	0.965	0.951	0.958	0.960	0.955	0.957	0.965	0.951	0.958	0.948	0.940	0.943
Politics	0.968	0.962	0.964	0.945	0.961	0.953	0.968	0.962	0.964	0.958	0.968	0.963
Science	0.949	0.943	0.945	0.922	0.933	0.926	0.949	0.943	0.945	0.945	0.937	0.941
Average	0.970	0.963	0.966	0.938	0.951	0.944	0.970	0.963	0.966	0.951	0.954	0.952

3.2 Qualitative Comparison

Table 4 shows representative naming examples contrasting OWNER-LLM and Span2Type-Ollama, where the two approaches differ most.

4 Analysis

4.1 OWNER-BERT = Span2Type-MLM

Both BERT-based methods achieve **identical** BERTScore across all domains (avg. $F_1 = 0.966$). This is expected: they use the same BERT-base-uncased MLM head with the same prompt template and voting mechanism on the same cluster assignments. The “**vegetarian**” artefact appears in both (8 of 92 clusters, 8.7%) for the same reason: ambiguous entities in the “‘X is a [MASK].’” pattern trigger BERT’s most frequent fill pattern from pretraining.

Table 4: Selected naming examples comparing LLM-based approaches. **Green** = correct/near-correct; **orange** = partially correct; **red** = incorrect.

Dom.	GT	OWNER-LLM	S2T-Ollama
AI	algorithm	Computer Science Concepts	artificial intelligence machine learning
AI	programlang	Programming languages	programming languages
AI	product	Robot	machine learning library
AI	person	Person	director
Lit.	book	Novel	character
Lit.	literarygenre	Literary work	novels
Lit.	event	Film Festivals	film festivals
Mus.	musicalartist	Singer	singer
Mus.	event	Event	multisport events
Mus.	album	Song	albums
Pol.	politicalparty	Political party	political party
Pol.	organisation	Government	legislative body
Pol.	person	Celebrity	actress
Sci.	scientist	Physicist	historical figures
Sci.	chemicalcompound	Chemical substance	chemical compound
Sci.	misc	Ethnic groups	biological processes

4.2 Span2Type-Ollama Outperforms OWNER-LLM

Span2Type-Ollama achieves a higher average F_1 (**0.952**) than OWNER-LLM (**0.944**), a gap of +0.008. The advantage comes from richer context: Span2Type-Ollama includes full example sentences in its prompt, allowing the LLM to infer fine-grained types such as *machine learning library* (GT: product), *legislative body* (GT: organisation), and *biological processes* (GT: misc).

OWNER-LLM, which only passes entity strings without their sentence context, is more susceptible to over-generalisation (e.g., *Celebrity* for **person**; *Stadiums* for **country**).

The improvement is most pronounced on **Literature** ($\Delta F_1 = +0.015$) and smallest on **Music** (-0.014), where Span2Type-Ollama over-specifies (*multisport events* for **event**).

4.3 Per-Domain Summary

- **AI:** BERT methods dominate ($F_1 = 0.989$). LLM methods struggle with abstract types like **algorithm** and **task**.
- **Literature:** Span2Type-Ollama achieves the highest F_1 (0.969), narrowly beating the BERT methods (0.974 vs 0.969 is within rounding; Span2Type-Ollama has higher Recall 0.970 vs 0.975).
- **Music:** All methods perform similarly (range 0.943–0.958). Both Ollama variants confuse **album** with song-related terms.
- **Politics:** BERT methods and Span2Type-Ollama are within 0.001 F_1 , all around 0.963–0.964.
- **Science:** BERT methods lead (0.945); Span2Type-Ollama is close (0.941). Fine-grained scientific types remain challenging for all approaches.

5 Conclusion

Across all four methods, BERT/MLM voting achieves the highest BERTScore ($F_1 = 0.966$) but produces single-token names and the *vegetarian* artefact. Among LLM-based approaches, **Span2Type-Ollama** ($F_1 = 0.952$) outperforms **OWNER-LLM** ($F_1 = 0.944$) by providing fuller sentence context to the model, yielding more precise multi-word names.

The ranking across all methods is:

$$\text{OWNER-BERT} = \text{Span2Type-MLM} (0.966) > \text{Span2Type-Ollama} (0.952) > \text{OWNER-LLM} (0.944)$$

For practical deployment where human-readable output matters, Span2Type-Ollama is recommended. For cost-free naming within an existing OWNER pipeline, the MLM voting approach remains the strongest baseline.

References

- [1] P.-Y. Genest, P.-E. Portier, E. Egyed-Zsigmond, and M. Lovisetto, “OWNER – Toward Unsupervised Open-World Named Entity Recognition,” *IEEE Access*, vol. 13, pp. 50077–50096, 2025.
- [2] T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, and Y. Artzi, “BERTScore: Evaluating Text Generation with BERT,” in *Proc. ICLR*, 2020.